

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotypes

Name: Rica Jane M. Torres	Instructor: Mr. Abner A. Bucol Mrs. Lilibeth A. Bucol
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Part I. Select a Human Disease and Gene

Tay-Sachs disease is a rare inherited disease that mainly affects the nervous system. It happens when the body does not have enough working β -hexosaminidase A (Hex A), an enzyme that helps break down GM2 ganglioside inside lysosomes. When the enzyme is missing or does not work properly, GM2 ganglioside starts to build up in nerve cells and causes damage. This can lead to problems such as delayed development, muscle weakness, seizures, and loss of vision and hearing (Lui et al., 2024).

The HEXA c.1274_1277dup is one mutation associated with Tay-Sachs disease. It involves the duplication of four nucleotides, TATC, in the HEXA coding sequence. This changes the reading frame and causes a frameshift, which changes the amino acids after the mutation and creates a premature stop codon. The mutation therefore produces a shorter HEXA protein that is expected to have little or no normal function. With reduced β -hexosaminidase A activity, GM2 ganglioside cannot be properly broken down and continues to accumulate in nerve cells (National Center for Biotechnology Information, 2026).

Part II. Research the Disease

A. Disease

13. What is the disease or phenotype?

- Tay-Sachs disease is a hereditary metabolic disorder that affects the nervous system. It is caused by mutations in the HEXA gene. The disease can have different forms, including infantile, juvenile, and adult-onset forms (National Center for Biotechnology Information (US), 1998).

14. What are its major clinical characteristics?

- It can cause progressive loss of motor skills, muscle weakness, seizures, vision and hearing loss, paralysis, and decline in mental functions. The symptoms vary depending on whether the disease begins during infancy, childhood, or adulthood (Mayo Foundation for Medical Education and Research, 2022).

15. Which cells, tissues, or organs are mainly affected?

- It mainly affects the nervous system, particularly the central and autonomic nervous

systems and the retina. GM2 gangliosides accumulate primarily in neurons, causing neuronal dysfunction and neurodegeneration. The brainstem, spinal cord, hippocampus, and retinal ganglion cells can also be affected (Lui et al., 2024).

16. What is its genetic basis?

- caused by mutations in both copies (alleles) of the HEXA gene, which is located on chromosome 15. These mutations result in reduced or absent activity of the β -hexosaminidase A enzyme (Lui et al., 2024).

17. What is its inheritance pattern, if applicable?

- It follows an autosomal recessive inheritance pattern, meaning that an affected individual inherits a disease-causing HEXA variant from both parents (Lui et al., 2024).

B. Gene and Normal Protein

18. What is the official gene symbol?

- The official gene symbol associated with Tay-Sachs disease is HEXA. The gene encodes the alpha subunit of β -hexosaminidase A (HexA), a lysosomal enzyme involved in the degradation of GM2 ganglioside.

19. On which human chromosome is the gene located?

- The HEXA gene is located on chromosome 15, specifically at chromosome region 15q23.

20. What does the gene normally encode?

- The HEXA gene encodes the alpha subunit of the lysosomal enzyme β -hexosaminidase A (HexA).

21. What is the normal biological function of the protein?

- The alpha subunit is part of β -hexosaminidase A, which normally helps break down GM2 ganglioside and other molecules containing terminal N-acetyl hexosamines. This process prevents GM2 from accumulating in cells, especially neurons.

22. Where in the cell is the protein normally found?

- The β -hexosaminidase A enzyme is normally found in the lysosomes, where it participates in the breakdown and recycling of cellular materials.

23. In what biological pathway or cellular process does it participate?

- The protein participates in lysosomal lipid degradation, specifically the breakdown of GM2 ganglioside. β -hexosaminidase A works together with the GM2 activator protein to degrade GM2 in lysosomes.

C. Documented Mutation

Gene	HEXA
Reference transcript	NM_000520.6
Exact variant notation	NM_000520.6(HEXA):c.1274_1277dup
Nucleotide change	4-bp duplication
Predicted protein change	p.Tyr427fs

Mutation type	Frameshift
ClinVar accession	VCV000003889.224
Clinical interpretation	Pathogenic
Associated disease	Tay-Sachs disease
Scientific reference	Myerowitz & Costigan, 1988 (PMID: 2848800)

Hypothesis

If a mutation is made in the normal HEXA coding sequence, it is expected to affect the HEXA protein depending on the type of mutation. The documented HEXA c.1274_1277dup mutation is expected to cause a frameshift, change the amino-acid sequence, and create a premature stop codon. On the other hand, the artificial three-nucleotide deletion is expected to remove one amino acid without changing the reading frame. These changes may affect the normal function of the HEXA protein and may be related to the development of Tay-Sachs disease.

Methods

This study used Galaxy to examine the effects of mutations in the HEXA gene. The normal HEXA coding sequence from NM_000520.6 was obtained from NCBI and uploaded to Galaxy as the Wild Type sequence. The Wild Type sequence was translated using the Transeq tool to get its predicted protein sequence. The sequence and protein length, reading frame, start and stop codons, and amino-acid sequence were recorded.

The documented HEXA c.1274_1277dup mutation was made using a copy of the Wild Type sequence by duplicating the TATC nucleotides at positions c.1274–c.1277. The mutated sequence was then translated using Transeq and compared with the Wild Type protein. The amino-acid changes, reading frame, protein length, and premature stop codon were checked.

For the artificial mutation, three nucleotides (GGC) were deleted from positions 71–73 of a copy of the Wild Type HEXA sequence. The sequence was then translated using Transeq. The WT, documented mutant, and artificial mutant sequences were compared to see the differences in mutation type, amino-acid sequence, reading frame, protein length, and premature stop codon. These results were used to understand how the different mutations affected the HEXA protein.

Part III. Obtain the Normal Reference Sequence

Reference transcript: NM_000520.6

CDS: 43(Start) –1632(End)

Reference protein: NP_000511.2

FASTA file: HEXA_WT_CDS.fasta

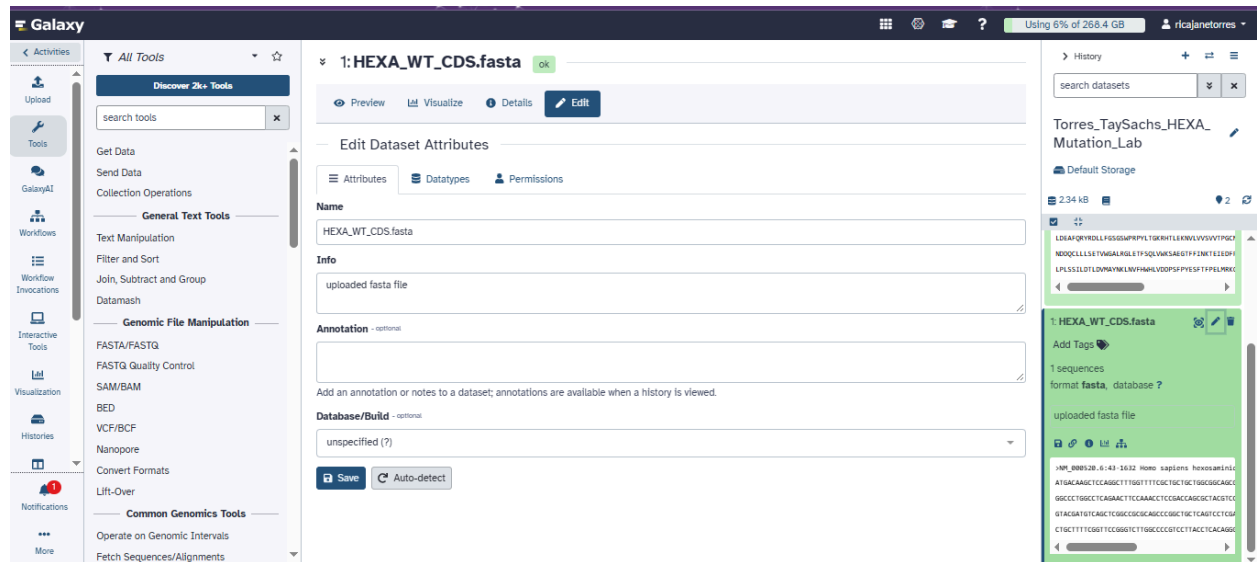


Figure 1. Renaming the wild-type HEXA coding sequence (CDS) to HEXA_WT_CDS.fasta in Galaxy prior to translating it into a protein sequence.

Part IV. Import the Normal Sequence into Galaxy

The wild-type HEXA coding sequence (CDS) was uploaded to Galaxy and used as the reference sequence for the analysis. A separate Galaxy history, Torres_TaySachs_HEXAMutation_Lab, was created to organize the sequences, datasets, and analysis results. The original WT sequence was kept unchanged and served as the baseline for comparing the documented and artificial mutations.

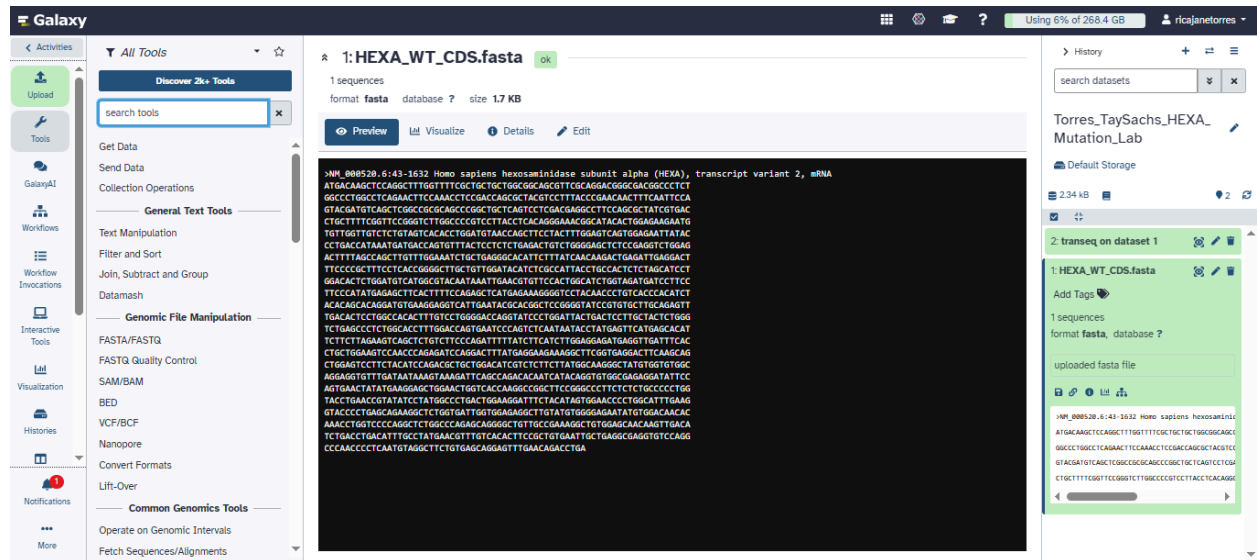


Figure 2. The normal HEXA coding sequence (WT CDS) was successfully uploaded to the Galaxy history Torres_TaySachs_HEXA_Mutation_Lab in FASTA format.

Part V. Establish the Wild-Type Control

CDS length	1,590 bp
Predicted protein length	529 amino acid
Start codon	ATG
Stop codon	TGA
Reading frame	Frame 1
First 10 amino acids	MTSSRLWFSL
Last 10 amino acids	GFCEQEFEQT

Part VI. Formulate a Mutation Hypothesis

Mutation and exact nucleotide change	NM_000520.6:c.1274_1277dup (TATC)
Number of nucleotide(s) affected	4 nucleotides
Predicted mutation type	Duplication
Predicted effect on the reading frame	Causes a frameshift because 4 nucleotides are added.

Predicted effect on protein length	Expected to produce a premature stop codon and a shorter protein.
Predicted effect on protein function	The mutation is expected to disrupt the HEXA protein, which may result in loss of normal enzyme function.

Part VII. Create the Mutant Sequence

Original nucleotide position(s)	c.1274–c.1277
Original sequence	TATC
Mutant sequence	TATCTATC
Number of bases inserted/duplicated	4 bases
Mutation type	Duplication
Mutant filename	HEXA_c1274_1277dup_CDS.fasta



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HEXA_WT_CDS.fasta - Notepad
File Edit Format View Help
>NM_009520.6:43-1632 Homo sapiens hexosaminidase subunit alpha (HEXA), transcript variant 2, mRNA
ATGACAAGCTCCAGGCTTTGGTTTTGCTGCTGCTGGCGGCAGCGTTTCGAGGACGGGCGACGGCCCTCT
GGCCCTGGCTCAGAACTTCAAACCTCCGACGAGCGCTACGTCCTTTACCGAACAACCTTCAATTCCA
GTACGATGTCAAGCTCGGCCGCGCAGGCCGGCTGCTCAGTCTCTGACGAGGCCCTTCCAGCGCTATCGTGAC
CTGCTTTTCGGTTCGGGCTTTGGCCCGTCTTACCTCACAGGGAACGGCATACTGGAGAAGAATG
TGTGGTTGTCTCTGTAGTCACACCTGGATGAACAGCTTCTCTATTTGGAGTCAGTGGAGAATTATAC
CTGACCATAAATGATGACCACTGTTTACTCTCTGAGACTGCTGGGAGCTCTCCGAGGCTCTGGAG
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TTCCCCGCTTCTCTACCGGGCTTGCTGTTGGATACATCTGCCATTACCTGCCACTCTCTAGCATCCT
GGACACTCTGGATGTCATGGCGTACAATAAATGAACGTGTTCCACTGGCATCTGGTAGTGATCCTTCC
TTCCATATGAGAGCTTCACTTTTCCAGAGCTCATGAGAAAGGGCTCTACAACCTGTCAACCATCT
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TGACACTCTGGCCACACTTTGCTCTGGGACAGGATCCCTGGATTACTGACTCTCTGTACTCTGGG
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TCTTCTAGAACTCAGCTCTGTCTTCCAGATTTTATCTTATCTTGGAGGAGATGAGGTTGATTTCAC
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CTGGAGTCTTTACATCCAGACGCTGCTGGACATCTCTCTTATGGAAGGGCTATGTTGGTGTGGC
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TACCTGAACCGTAATATCTATGCTGCTGAGTGAAGGATTTACATAGTGAACCCCTGGCATTTGAAG
GTACCCCTGAGCAGAAGGCTCTGGTGTGTTGGAGAGGCTTGTATGTGGGAGAATATGTGACAAAC
AAACCTGGTCCCAGGCTCTGGCCAGAGCAGGGGCTGTTGCCGAAAGGCTGTGGAGCAACAAGTTGACA
TCTGACCTGACATTGCTATGAACGTTTGTACACTTCCGCTGTGAATTGCTGAGGCGAGGTGTCAGG
CCCCAACCCCTCAATGTAGGCTTCTGTGAGCAGGAGTTTGAACAGACCTGA

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Figure 3. Manually edited HEXA coding sequence showing the duplication of the TATC nucleotide sequence to reproduce the documented c.1274_1277dup mutation.

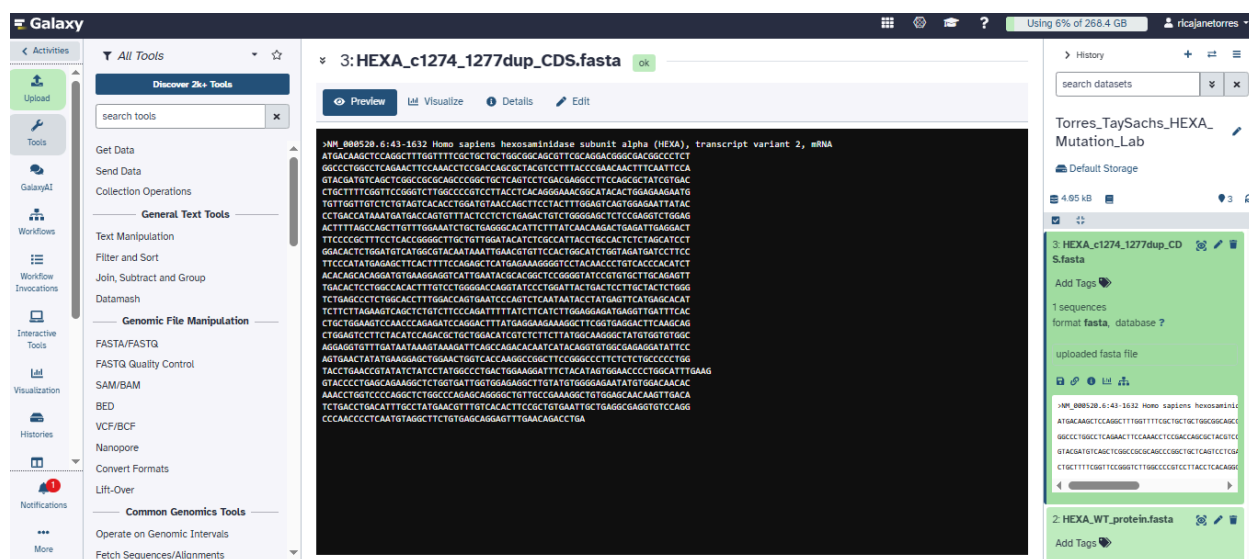


Figure 4. Mutant HEXA CDS sequence uploaded to Galaxy for subsequent analysis.

Part VIII. Translate the Mutant Sequence

Mutant CDS Length	1,594 bp
Mutant protein length	431 amino acids
Reading frame	Frame 1
Location of the first amino acid difference	Amino acid 427
Presence or absence of a premature stop codon	Present
Approximate number of amino acids affected.	5 amino acids after the frameshift

In this table shows that the documented HEXA c.1274_1277dup mutation increased the CDS length from the normal 1,590 bp to 1,594 bp. The four-nucleotide duplication caused a frameshift, with the first amino-acid difference appearing at amino acid 427. A premature stop codon was also produced, resulting in a shorter predicted protein of 431 amino acids. Overall, the mutation changes the downstream protein sequence and may affect the normal function of HEXA.

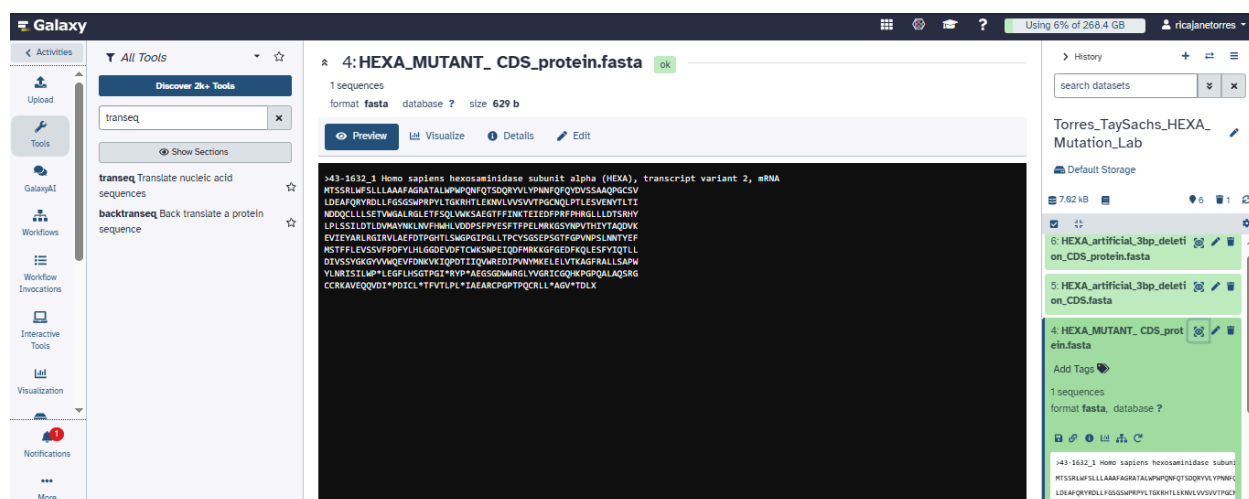


Figure 5. Translation of the mutant HEXA coding sequence containing the c.1274_1277dup mutation using Galaxy TranslateSeq, showing the predicted mutant protein sequence and premature stop codon.

Part IX. Compare WT and Mutant Proteins

At what amino-acid position do the sequences first differ?	Amino acid 427
Is only one amino acid affected?	No
Are multiple downstream amino acids changed?	Yes. The frameshift changes the downstream amino-acid sequence
Was an amino acid deleted or inserted?	No single amino acid was directly deleted or inserted. Instead, 4 nucleotides (TATC) were duplicated, causing a frameshift
Was a premature stop codon produced?	Yes
Did the reading frame change?	Yes. A 4-nucleotide duplication causes a frameshift.
Did the protein length change?	Yes. The WT protein is 529 aa, while the predicted mutant protein is approximately 431 aa.
Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?	Frameshift mutation

In this table, it shows that the documented HEXA c.1274_1277dup mutation causes a major change in the protein. The sequences are the same up to amino acid 426, but after the duplication at least the following amino acids are changed because of the frameshift. A premature stop codon is also produced, making the mutant protein shorter (431 aa) than the WT protein (529 aa). This suggests that the mutation may greatly affect the normal structure and function of the HEXA protein.

Part X. Explain the Molecular Consequence

HEXA c.1274_1277dup → TATC is duplicated → frameshift occurs → amino acids after position 427 are changed → premature stop codon → shorter HEXA protein → reduced β -hexosaminidase A function → GM2 ganglioside builds up in nerve cells → nerve cell damage → Tay-Sachs disease.

The c.1274_1277dup mutation in the HEXA gene involves the duplication of four nucleotides, TATC. Since four bases were added, the reading frame changed and caused a frameshift in the coding sequence. Based on the Galaxy translation, the mutant protein first differs from the normal protein at about amino acid 427 and contains a premature stop codon. As a result, the mutant protein is shorter than the normal 529-amino-acid HEXA protein. The altered protein sequence may disrupt its normal structure and function, reducing the activity of β -hexosaminidase A. The documented variant is also predicted to cause protein truncation or nonsense-mediated mRNA decay, which can lead to little or no functional enzyme

According to Lui et al. (2024), HEXA forms part of β -hexosaminidase A, an enzyme found in lysosomes that helps break down GM2 ganglioside. When the enzyme is deficient or does not function properly, GM2 ganglioside cannot be properly broken down and accumulates in nerve cells. This buildup can damage neurons and contribute to the progressive neurological problems associated with Tay-Sachs disease.

Part XI. Second Experiment: Create Your Own Mutation

Artificial Mutation	Three-nucleotide deletion
Nucleotide Position	c.71–c.73 (GGC)
Original Sequence	GGC
Mutant Sequence	GGC deleted
Number of bases affected	3 bases
Predicted Mutation Type	Deletion

Predicted Effect in Reading Frame	No frameshift; the reading frame remains Frame 1.
Predicted Effect on Protein	Expected to remove one amino acid while keeping the downstream amino-acid sequence in the same reading frame

In the artificial mutation, I deleted three nucleotides (GGC) from the HEXA coding sequence at positions c.71–c.73. Since three nucleotides were removed, the reading frame is expected to remain unchanged because three nucleotides make up one codon. The predicted protein is expected to lose one amino acid, while the amino acids after the deletion should remain in the same reading frame. Compared with the documented mutation, this artificial mutation is expected to have a smaller effect on the HEXA protein.

The screenshot displays the Galaxy web interface. On the left, a sidebar lists various tools and workflows. The main area shows a dataset named '5: HEXA_artificial_3bp_deletion_CDS.fasta' with a preview of its FASTA content. The sequence is a single-line FASTA entry for a hexosaminidase subunit alpha (HEXA) transcript variant 2. The right sidebar shows the history of datasets, including 'Torres_TaySachs_HEXA_Mutation_Lab' and '5: HEXA_artificial_3bp_deletion_CDS.fasta'.

Figure 6. The artificial HEXA CDS containing the three-nucleotide deletion was successfully uploaded to the Galaxy history Torres_TaySachs_HEXA_Mutation_Lab in FASTA format.

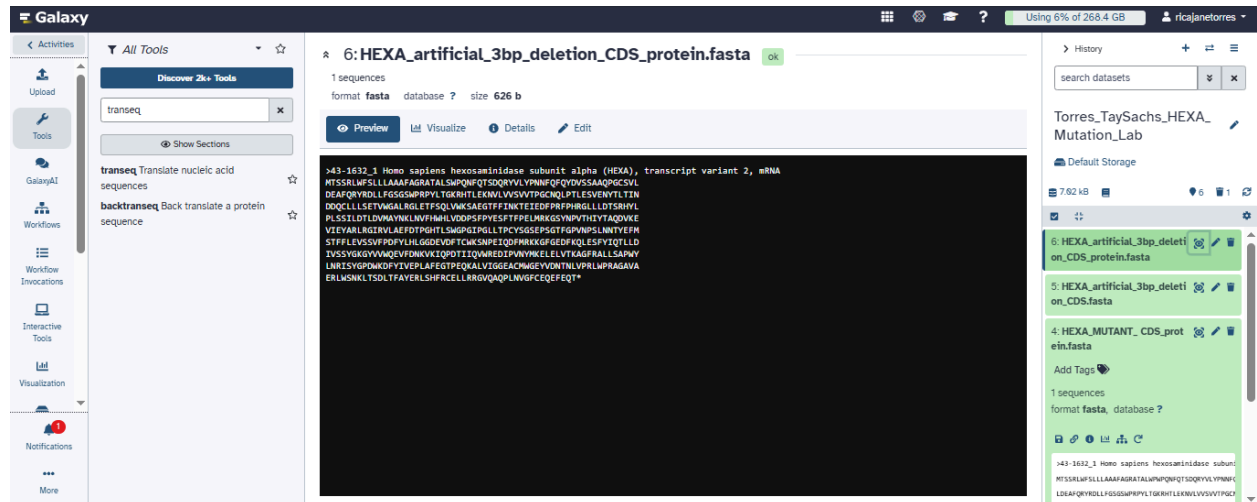


Figure 7. Translation of the artificial HEXA CDS containing the three-nucleotide deletion using Galaxy TranslateSeq.

Part XII. Compare the Documented and Artificial Mutations

	WT Sequence	Documented mutation	Artificial mutation
CDS Length	1,590 bp	1,594 bp	1,587 bp
Protein Length	529 aa	431 aa	528 aa
Mutation Type	None	4-bp duplication / frameshift	3-bp deletion / in-frame deletion
Reading frame changed	No, Frame 1	Yes, frameshift	No, remains Frame 1
Premature stop codon appeared	No	Yes	No, expected
Amino acids affected	None	Begins at amino acid 427; downstream sequence is altered and protein is truncated	One amino acid deleted
Expected functional consequence	Normal HEXA function	Likely loss or major reduction of functional HEXA protein	Likely a small structural change; most of the protein sequence remains unchanged

This table shows the documented and artificial mutations had different effects on the HEXA protein. The documented mutation adds four nucleotides, causing a frameshift and premature stop codon, which makes the predicted protein much shorter. In contrast, the artificial mutation removes three nucleotides, so the reading frame stays the same and only one amino acid is expected to be removed. This shows that the number of nucleotides changed can affect how much a mutation changes the protein.

Part XIII. Interpretation Questions

32. Why does the exact location of a mutation matter?

- The exact location of a mutation matters because it determines which part of the gene or protein will be affected. A mutation in a coding region can change the protein sequence and may affect its function. In our experiment, the HEXA c.1274_1277dup mutation causes a frameshift starting at about amino acid 427 and results in a premature stop codon.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- Deleting three nucleotides can remove one complete codon without changing the reading frame because one codon has three nucleotides. On the other hand, deleting one or two nucleotides causes a frameshift. This changes the way the following nucleotides are read and can change many amino acids.

34. Does every mutation change the amino-acid sequence? Explain.

- No, not every mutation changes the amino-acid sequence. Some mutations are synonymous mutations, where the changed codon still codes for the same amino acid. Mutations outside the coding region can also affect the gene without directly changing the protein sequence.

35. Does every amino-acid substitution destroy protein function? Explain.

- No, an amino-acid substitution does not always destroy protein function. The effect depends on where the amino acid is located and what amino acid replaces it. Some changes may have little effect, while others can affect the structure or function of the protein. In our HEXA mutation, the frameshift changes the amino-acid sequence and produces a premature stop codon.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- A frameshift can affect many amino acids because deleting one nucleotide changes how the nucleotides are grouped into codons. Because of this, the codons after the mutation are read differently, which can change many of the following amino acids and may also create a premature stop codon.

37. Why might a premature stop codon produce a nonfunctional protein?

- A premature stop codon stops protein production too early. This produces a shorter protein that may be missing important parts needed for its normal function. Because of this, the protein may not work properly.

38. Could a mutation affect protein function without greatly changing protein length?

- Yes. A mutation can change one amino acid without changing the overall length of the protein. For example, a missense mutation can change an important amino acid and affect the protein's structure or function even though its length stays the same.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes. A mutation can cause disease without changing the protein sequence if it affects how much protein is produced. For example, a mutation in a promoter or regulatory region can reduce gene expression. A mutation affecting RNA splicing can also change how the RNA is processed.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- Our analysis showed that the HEXA c.1274_1277dup mutation adds four nucleotides, TATC, which causes a frameshift. The mutant protein first differs from the normal protein at about amino acid 427 and has a premature stop codon. The mutant protein is also shorter than the normal protein. ClinVar classifies this variant as pathogenic and associates it with Tay-Sachs disease.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- Our computational results directly show the DNA change, frameshift, amino-acid changes, premature stop codon, and predicted protein length. However, our analysis cannot directly prove that the mutant protein is nonfunctional or that it causes GM2 ganglioside to accumulate in nerve cells. These effects need published experimental and clinical evidence. Studies show that reduced or absent β -hexosaminidase A activity

causes GM2 ganglioside to accumulate in nerve cells, which is associated with Tay-Sachs disease.

Part XV. GitHub Commit History

Github link: <https://github.com/torresricajane/TaySachs-HEXA-Mutation-Lab>

Galaxy link: <https://usegalaxy.org/u/ricajanetorres/h/torres-taysachs-hexa-mutation-lab>

Part XVI. Final Report

Limitations

This study used Galaxy to predict how the HEXA mutations could affect the protein sequence. However, the analysis was based only on DNA sequence and translation, so it could not directly show whether the mutant HEXA protein is actually produced, folded correctly, or still works normally. The analysis also did not directly measure β -hexosaminidase A activity or GM2 ganglioside accumulation in cells. The artificial mutation was also created only for comparison and is not necessarily a disease-causing mutation. Therefore, the results mainly show the predicted effects of the mutations and should be supported by experimental and clinical evidence.

Conclusion

This activity showed how a change in the DNA sequence can affect the protein produced by a gene. In the HEXA c.1274_1277dup mutation, four nucleotides, TATC, were duplicated, causing a frameshift and a premature stop codon. The predicted mutant protein was shorter than the normal HEXA protein, with the first difference occurring around amino acid 427. In comparison, the artificial three-nucleotide deletion was expected to keep the reading frame and remove only one amino acid. Overall, the activity demonstrated that the type and number of nucleotides changed can affect the protein sequence and its possible function. The results also helped connect the HEXA mutation to the reduced function of β -hexosaminidase A and the development of Tay-Sachs disease.

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